Probabilistic Collision Cone Control Barrier Function

Maciej Różewicz

September 14, 2025

Abstract

Safety is a fundamental requirement for autonomous systems operating in dynamic and uncertain environments. Control Barrier Functions (CBFs) have emerged as a powerful tool for enforcing safety constraints in real time, but most existing approaches assume perfect state information, which is rarely available in practice due to sensor noise and perception errors. This paper introduces the Probabilistic Collision Cone Control Barrier Function (PCC-CBF), a novel framework that explicitly accounts for uncertainty in the relative position and velocity of obstacles by modeling them as Gaussian random variables. The proposed method reformulates the classical collision cone condition as a chance constraint, ensuring that the probability of collision remains below a specified risk threshold. We derive tractable conditions for chance-constrained safety and demonstrate the effectiveness of the PCC-CBF approach through simulations on unicycle and bicycle models, as well as preliminary real-world experiments.

1 Introduction

Ensuring safety is a fundamental requirement in the design of autonomous systems, particularly in the context of mobile robots and self-driving vehicles operating in uncertain and dynamic environments. When interacting with other traffic participants or moving obstacles, collision avoidance must be guaranteed without compromising the ability to reach a desired goal.

Among various approaches to safety-critical control, *Control Barrier Functions* (CBFs) have emerged as a powerful tool for enforcing forward invariance of safe sets in control-affine systems [2]. CBFs provide a formal mechanism for encoding safety constraints that can be enforced through online quadratic programming while maintaining system performance. However, the majority of existing CBF-based methods assume deterministic knowledge of the state and environment, which is rarely available in practice due to sensor noise and imperfect perception.

For collision avoidance, the notion of the *Collision Cone* (CC) has been widely used to characterize dangerous relative motions between agents [4]. The CC provides a geometric condition to determine whether two objects are on a collision course, and several recent works have combined this concept with CBFs to ensure collision-free navigation [5]. Yet, these approaches still rely on deterministic estimates of relative position and velocity, making them sensitive to uncertainty. As perception in real-world scenarios is inherently noisy, such methods may yield overly conservative or unsafe behavior.

To address this limitation, in this work we propose a *Probabilistic Collision Cone Control Barrier Function (PCC-CBF)*. The key idea is to extend the deterministic CC-CBF formulation by explicitly modeling uncertainty in relative position and velocity as Gaussian random variables. This allows the safety condition to be reformulated as a chance constraint, ensuring that the probability of collision remains below a specified risk threshold. The resulting probabilistic safety certificates can be integrated into a quadratic programming framework for real-time safe control.

The main contributions of this paper are threefold:

- We introduce a probabilistic formulation of Collision Cone Control Barrier Functions that accounts for uncertainty in perception.
- We derive tractable conditions for chance-constrained safety by exploiting Gaussian approximations of state uncertainty.
- We demonstrate the effectiveness of the proposed PCC-CBF approach in simulation studies on unicycle and bicycle models, as well as preliminary real-world experiments.

The remainder of this paper is organized as follows: Section 2 reviews related work. Section 3 introduces the classical CBF framework and its integration with collision cones. Section 4 presents the proposed probabilistic extension. Simulation and experimental results are shown in Section 5, followed by conclusions in Section 6.

2 Literature Review

The literature on safety-critical control and collision avoidance in autonomous systems is extensive and rapidly evolving. This section reviews key developments relevant to Control Barrier Functions (CBFs), collision cone methods, and probabilistic safety guarantees.

Control Barrier Functions (CBFs): CBFs have become a foundational tool for enforcing safety constraints in real-time control of nonlinear systems. The seminal work by **(author?)** [2] ("Control Barrier Function Methods for Safety Critical Systems") formalized the use of CBFs for forward invariance of safe sets, enabling their integration into quadratic programming frameworks for safety-critical control. Extensions have addressed adaptive systems, input constraints, and multi-agent scenarios [1] ("Control Barrier Functions: Theory and Applications") and [**?**] ("Exponential Control Barrier Functions for Enforcing High Relative-Degree Safety-Critical Constraints").

Collision Cone Methods: The collision cone (CC) concept, originally introduced for spacecraft and robotics applications, provides a geometric framework for predicting potential collisions based on relative positions and velocities [**?**] ("Collision cones for motion planning and collision avoidance"). Recent works have combined CCs with CBFs to create less conservative and more dynamically aware safety constraints [5] ("Observer-Based Collision Cone Control Barrier Functions for Multi-Agent Systems") and [3] ("Collision Cone Approach Control Barrier Functions for Multi-Agent Collision Avoidance"). These approaches have demonstrated improved performance in dynamic and multi-agent environments.

Probabilistic Safety and Uncertainty: Most classical CBF and CC-based methods assume perfect state information, which is unrealistic in practice. To address this, several studies have explored robust and probabilistic extensions. Chance-constrained CBFs [**?**] ("Probabilistic Safety for Control Barrier Functions with Applications to Adaptive Cruise Control") and [**?**] ("Chance-Constrained Control Barrier Functions for Safety-Critical Control under Uncertainty"), as well as stochastic safety certificates [**?**] ("Guaranteeing Safety of Learning-Based Control Policies in Unknown Environments"), have been proposed to account for sensor noise and estimation errors. Gaussian approximations and linearization techniques are commonly used to derive tractable safety conditions under uncertainty.

Applications: CBF and CC-based safety frameworks have been applied to a wide range of robotic systems, including mobile robots, autonomous vehicles, and aerial drones [1] ("Control Barrier Functions: Theory and Applications") and [5] ("Observer-Based Collision Cone Control Barrier Functions for Multi-Agent Systems"). The integration of probabilistic safety guarantees is particularly important for real-world deployment, where perception uncertainty and dynamic obstacles are prevalent.

In summary, while CBFs and collision cone methods provide powerful tools for safety-critical control, explicitly accounting for uncertainty remains an active area of research. The proposed Probabilistic Collision Cone Control Barrier Function (PCC-CBF) builds on these foundations to offer a tractable and robust approach for safe navigation in uncertain environments.

3 Control Barrier Function

Control Barrier Functions (CBFs) are mathematical constructs used in control theory to ensure the safety of dynamical systems. They provide a systematic way to enforce safety constraints by defining a safe set of states that the system should remain within. CBFs are particularly useful in scenarios where safety is critical, such as in autonomous vehicles, robotics, and aerospace applications. A Control Barrier Function $h(x)$ is a scalar function that maps the state space to the real numbers. The safe set is defined as the set of states where $h(x) \geq 0$. The boundary of the safe set is given by $h(x) = 0$, and states where $h(x) < 0$ are considered unsafe. To ensure that the system remains within the safe set, the CBF must satisfy certain conditions. Specifically, for a dynamical system described by equation (1):

$$\dot{x} = f(x) + g(x)u. \tag{1}$$

Where x is the state, u is the control input, and $f(x)$ and $g(x)$ are system dynamics, the CBF must satisfy the following inequality (2):

$$\dot{h}(x, u) + \alpha(h(x)) \geq 0 \tag{2}$$

Specifying the barrier function $h(x)$ is a critical design choice, as it determines the shape of the safe set. In many robotic applications, simple distance-based formulations (e.g., maintaining $||p_r|| \geq r$ where r is the minimum separation distance) are used to enforce collision avoidance. While intuitive, such constraints are often overly conservative: they do not account for the relative motion between agents, and may unnecessarily restrict the system's ability to navigate safely in dense or dynamic environments.

To address this limitation, the concept of the *Collision Cone* (CC) has been introduced as a geometric tool to characterize dangerous relative velocities. The CC describes the set of relative velocity vectors that would eventually lead to a collision, given the current relative position between two agents. Incorporating the CC into the barrier function allows one to explicitly capture the interplay between position and velocity, resulting in less conservative and more dynamically appropriate safety constraints.

This motivates the definition of a Collision Cone Control Barrier Function (CC-CBF), which encodes the requirement that the relative motion of the ego vehicle and obstacle remains outside the collision cone. The corresponding formulation is presented below.

Collision cone:

$$h(x) = < p_r, v_r > + ||p_r|| \cdot ||v_r|| \cos\phi \tag{3}$$

where $\cos\phi = \frac{\sqrt{||p_r||^2 - r^2}}{||p_r||}$

[3]

The collision cone is a geometric concept used to predict potential collisions between two moving objects, such as vehicles or robots. It defines the set of relative velocity vectors that, if maintained, will eventually result in a collision given the current positions of the objects.

In more detail:

Imagine two agents: an ego vehicle and an obstacle. The relative position between them is p_r, and the relative velocity is v_r. The collision cone is constructed by drawing lines from the ego vehicle to the edges of the obstacle, forming a cone-shaped region in velocity space. If the relative velocity v_r lies inside this cone, the ego vehicle is on a collision course with the obstacle. If v_r is outside the cone, the agents will not collide, assuming their velocities remain constant. By incorporating the collision cone into a control barrier function, the system can enforce safety constraints that are sensitive to both position and velocity, allowing for less conservative and more flexible collision avoidance compared to simple distance-based rules. This approach is especially useful in dynamic environments where both agents may be moving.

Derivative of such defeined barrier function has following form:

$$\begin{aligned}\dot{h}(x,u) =& < \dot{p}_r, v_r > + < p_r, \dot{v}_r > \\ &+ < v_r, \dot{v}_r > \frac{\sqrt{||p_r||^2 - r^2}}{||v_r||} \\ &+ < p_r, \dot{p}_r > \frac{||p_r||}{\sqrt{||p_r||^2 - r^2}}\end{aligned} \tag{4}$$

This formula will be formulated in next section to has form as in equaltion 2.

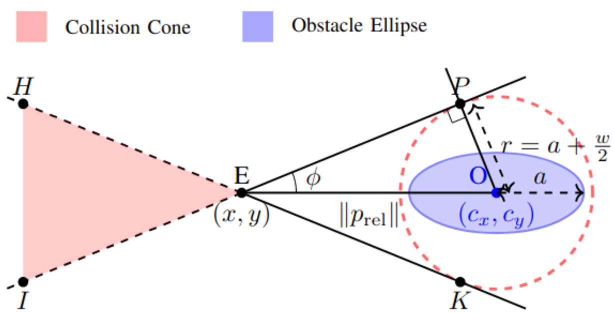

Figure 1: Construction of collision cone for an alliptical obstacle considering the ego vehicle's dimensions (width w).

4 Probabilistic Collision Cone CBF

The Collision Cone Control Barrier Function (CC-CBF) introduced in the previous section provides a deterministic safety certificate. In particular, the condition (2) ensures that the relative motion between the robot and an obstacle remains outside the collision cone. However, this formulation implicitly assumes perfect knowledge of the obstacle's position and velocity. In real-world scenarios, perception is subject to sensor noise, occlusions, and state estimation errors, which may lead to unsafe behavior if not explicitly accounted for.

To incorporate uncertainty, we model the relative position p_r and velocity v_r of the obstacle as random variables. Let

$$p_r = \hat{p}_r + \delta_p, \qquad v_r = \hat{v}_r + \delta_v, \tag{5}$$

where $\hat{p}_r, \hat{v}_r$ denote the estimated values, and δ_p, δ_v are zero-mean Gaussian random variables with covariances Σ_p and Σ_v, respectively:

$$\delta_p \sim \mathcal{N}(0, \Sigma_p), \qquad \delta_v \sim \mathcal{N}(0, \Sigma_v). \tag{6}$$

By linearizing the deterministic barrier function around the estimated state $\hat{x}$, we obtain a first-order approximation:

$$h(x) \approx h(\hat{x}) + \nabla_p h(\hat{x})^T \delta_p + \nabla_v h(\hat{x})^T \delta_v \tag{7}$$

where:

- $\delta_p \sim \mathcal{N}(0, \Sigma_p)$ and $\delta_v \sim \mathcal{N}(0, \Sigma_v)$,
- $\nabla_p h(\hat{x})^T = v_r^T + \frac{||v_r||}{\Delta} p_r^T$,
- $\nabla_v h(\hat{x})^T = p_r^T + \frac{\Delta}{||v_r||} v_r^T$,
- $\Delta = \sqrt{||p_r||^2 - r^2}$.

Similarly, the derivative $\dot{h}(x,u)$ can be expressed as:

$$\begin{gathered}\dot{h}(x,u) \approx \dot{h}(\hat{x},u) \\ + \nabla_p \dot{h}(\hat{x},u)^T \delta_p + \nabla_v \dot{h}(\hat{x},u)^T \delta_v + \nabla_a \dot{h}(\hat{x},u)^T \delta_a,\end{gathered}$$

where:

- $\nabla_p \dot{h}(\hat{x},u)^T = a_r^T + \frac{<v_r,a_r>}{||v_r||} \frac{p_r}{\Delta} + \frac{||v_r||}{\Delta} v^T - \frac{||v_r||}{\Delta^3} < p_r, v_r > p_r$,
- $\nabla_v \dot{h}(\hat{x},u)^T = 2v_r + \Delta \left(\frac{a_r}{||v_r||} - \frac{<v_r,a_r>}{||v_r||^3} v_r \right) + \frac{1}{\Delta} \left(||v_r|| p_r + \frac{<p_r,v_r>}{||v_r||} v_r \right)$,
- $\nabla_a \dot{h}(\hat{x},u)^T = p_r + \frac{\Delta}{||v_r||} v_r$.

This allows us to represent the safety constraint (2) as a random variable $c_h(x,u)$:

$$\overbrace{\underbrace{\dot{h}(\dot{x},u) + \alpha(h(\dot{x},u))}_{\mu_{c_h}} + \delta_{c_h}}^{c_h(x,u)} \geq 0, \tag{8}$$

where:

- μ_{c_h} is the mean of $c_h(x,u)$ random variable,
- δ_{c_h} is a zero-mean Gaussian random variable with variance $\sigma_{c_h}^2 = \left(\nabla_p \dot{h}(\hat{x},u) + \nabla_p h(\hat{x}) \right)^T \delta_p + \left(\nabla_v \dot{h}(\hat{x},u) + \nabla_v h(\hat{x}) \right)^T \delta_v + \nabla_a \dot{h}(\hat{x},u)^T \delta_a$.

Instead of requiring $c_h(x,u) \geq 0$ deterministically, we impose a chance constraint:

$$\mathbb{P}\left[c_h(x,u) \geq 0\right] \geq 1-\nu, \tag{9}$$

where $\nu \in (0,1)$ denotes the acceptable risk level. Under the Gaussian approximation, condition (9) is equivalent to

$$\mu_{c_h} - \Phi^{-1}(1-\nu)\sigma_{c_h} \geq 0, \tag{10}$$

where Φ^{-1} is the inverse standard normal cumulative distribution function. This yields a tractable inequality constraint that can be incorporated into the quadratic program used for CBF-based control.

Equation (10) constitutes the proposed *Probabilistic Collision Cone Control Barrier Function (PCC-CBF)*. It ensures that the system maintains safety with probability at least $1-\nu$, while explicitly accounting for uncertainty in perception.

5 Experiments

Experiments were planned in two steps: first simulations - implemented in MATLAB Autonomous Driving Toolbox, second on real robots.

5.1 Motion Models

Two types of robot models are commonly encountered in industrial practice. The first is the unicycle model, typically used for indoor mobile robots. The second is the bicycle model, which serves as a standard representation for cars in driver assistance systems and autonomous driving applications.

5.1.1 Unicycle Model

The unicycle kinematic model is a simple yet effective representation for differential-drive robots and other mobile platforms that can move forward and rotate, but cannot move sideways. The model captures the essential nonholonomic constraint that the robot's velocity is always aligned with its heading direction.

In this model, the state vector consists of the position (x_p, y_p) of the robot in the plane, its orientation θ, linear velocity v, and angular velocity ω. The control inputs are the linear acceleration a and angular acceleration δ.

The continuous-time kinematic equations for the unicycle model are:

$$\underbrace{\begin{bmatrix} \dot{x}_u \\ \dot{y}_u \\ \dot{\theta}_u \\ \dot{v}_u \\ \dot{\omega}_u \end{bmatrix}}_{\dot{x}_u} = \underbrace{\begin{bmatrix} v_u \cos\theta_u \\ v_u \sin\theta_u \\ \omega_u \\ 0 \\ 0 \end{bmatrix}}_{f_u(x_u)} + \underbrace{\begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}}_{g_u(x_u)} \underbrace{\begin{bmatrix} a_u \\ \delta_u \end{bmatrix}}_{u_u}. \tag{11}$$

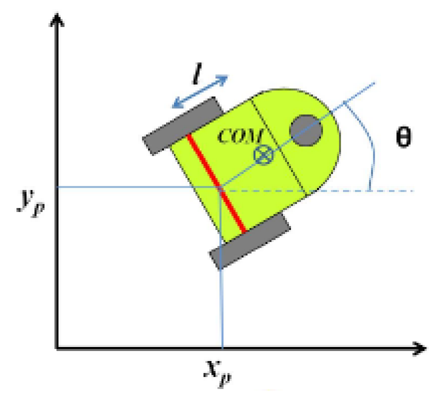

Figure 2: Schematic of unicycle model.

State: The state vector is given by:

$$x_u = \begin{bmatrix} x_u & y_u & \theta_u & v_u & \omega_u \end{bmatrix}^T,$$

where (x_u, y_u) represents the position, θ_u the heading angle, v_u the linear velocity, and ω_u the angular velocity.

Control: The control input vector is given by:

$$u_u = \begin{bmatrix} a_u & \delta_u \end{bmatrix}^T,$$

where a_u is the linear acceleration and δ_u is the angular acceleration.

This model is widely used in robotics for motion planning and control due to its simplicity and ability to capture the key motion constraints of many ground vehicles.

5.1.2 Bicycle Model Model

The bicycle model is a widely used kinematic representation for car-like vehicles, capturing the essential nonholonomic constraints and steering dynamics. In this model, the vehicle is abstracted as a single-track system with a front steering wheel and a rear driving wheel. The state vector typically includes the position (x_p, y_p) of the vehicle's reference point (often the rear axle midpoint), the heading angle θ, the longitudinal velocity v, and the yaw rate ω. The control inputs are the longitudinal acceleration a and the steering angle rate δ.

The continuous-time kinematic bicycle model is given by:

$$\underbrace{\begin{bmatrix} \dot{x}_b \\ \dot{y}_b \\ \dot{\theta}_b \\ \dot{v}_b \end{bmatrix}}_{\dot{x}_b} = \underbrace{\begin{bmatrix} v\cos\theta \\ v\sin\theta \\ 0 \\ 0 \end{bmatrix}}_{f_b(x_b)} + \underbrace{\begin{bmatrix} 0 & -v_b\sin\theta_b \\ 0 & -v_b\cos\theta_b \\ 0 & \frac{v_b}{L} \\ 1 & 0 \end{bmatrix}}_{g_b(x_b)} \underbrace{\begin{bmatrix} a_b \\ \delta_b \end{bmatrix}}_{u_b} \tag{12}$$

where L is the wheelbase (distance between front and rear axles).

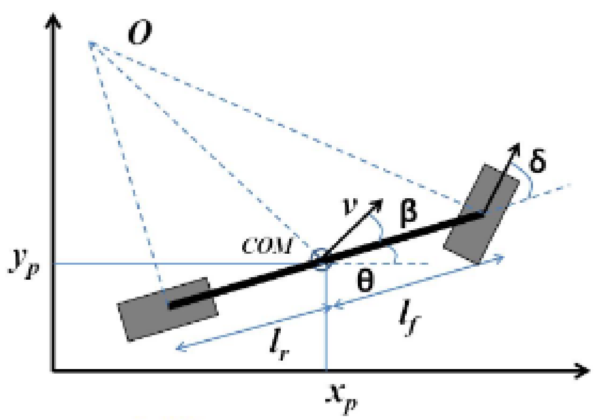

Figure 3: Schematic of bicycle model.

State: The state vector is given by:

$$x_b = \begin{bmatrix} x_b & y_b & \theta_b & v_b \end{bmatrix}^T,$$

where (x_b, y_b) represents the position, θ_b the heading angle, and v_b the longitudinal velocity.

Control: The control input vector is given by:

$$u_b = \begin{bmatrix} a_b & \delta_b \end{bmatrix}^T,$$

where a_b is the longitudinal acceleration and δ_b is the steering angle rate.

This model captures the coupling between steering and heading, and is commonly used for path planning and control in autonomous driving applications.

5.2 Simulations

MATLAB, Autonomous Driving

5.3 Real Experiments

6 Summary

References

[1] A. D. Ames, S. Coogan, M. Egerstedt, G. Notomista, K. Sreenath, and P. Tabuada. Control barrier functions: Theory and applications. In *2019 18th European Control Conference (ECC)*, pages 3420–3431, 2019.

[2] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada. Control barrier function based quadratic programs for safety critical systems. *IEEE Transactions on Automatic Control*, 62(8):3861–3876, 2017.

[3] M. Tayal, B. G. Goswami, K. Rajgopal, R. Singh, T. Rao, J. Keshavan, P. Jagtap, and S. Kolathaya. A collision cone approach for control barrier functions, 2024.

[4] P. Thontepu, B. G. Goswami, M. Tayal, N. Singh, S. S. P. I, S. S. M. G, S. Sundaram, V. Katewa, and S. Kolathaya. Collision cone control barrier functions for kinematic obstacle avoidance in ugvs. arXiv preprint arXiv:2209.11524, 2022.

[5] Y. Wang and X. Xu. Observer-based control barrier functions for safety critical systems. In *2022 American Control Conference (ACC)*, pages 709–714, 2022.